Muhammad Ibtihaj Tahir

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 4+ years across computer vision, financial ML, and audio, now focused on LLMs and agentic AI. Delivered real-time multi-task vision inference on live IP cameras; recently built and launched a production app end-to-end through an agentic Claude Code workflow, and currently implementing LLM transformer components from scratch on a custom AMD ROCm stack. Seeking AI Engineer roles spanning the full lifecycle, from data ingestion and feature engineering through training, deployment, and monitoring. I am as comfortable building AI systems as building with them.

SKILLS

- **Languages:** Python, C++, Swift, SQL
- **ML & AI:** PyTorch, TensorFlow, scikit-learn, OpenCV, NLTK, word2vec
- **LLM & Agents:** Ollama, Open WebUI, OpenClaw
- **MLOps & Infra:** Docker, Linux, Git, AWS EC2, Google Cloud
- **Data & Visualisation:** Pandas, NumPy, Matplotlib, Seaborn, Power BI
- **Edge & Hardware:** Raspberry Pi, Intel NCS, Core ML, Apple Accelerate, Arduino Uno
- **Professional:** Stakeholder presentations, staff training & mentoring, remote cross-timezone collaboration, customer-facing communication

EXPERIENCE

Currys — **Chester, England**
Sales Colleague | Data & Analytics Projects — *June 2024 - Present*

- **Headset Accountability System**: Built and launched a Microsoft Power App for scan-based check-out and check-in of shared VoCoVo headsets, recording who, when, and where for every device movement in a SharePoint-list backend (chosen over Dataverse to avoid premium licensing), giving managers live availability and a full audit trail after 4 devices went missing untraced.
- **AI-Assisted Engineering**: Delivered the app end-to-end via an agentic coding workflow with Claude Code, applying context-engineering practices, structured documentation handoffs between sessions, to keep a long multi-session build consistent and grounded within model context limits.
- **NLP Customer Insights**: Built an end-to-end NLP pipeline to process raw CSAT feedback, applying sentiment analysis and generating colleague-level Word Clouds within a custom Power BI dashboard; presented to management as a self-initiated analytics prototype.
- **Sales Forecasting**: Developed a Polynomial Regression model with confidence intervals to forecast individual sales performance trends; independently presented to management as an exploratory data analytics initiative.

Green Canvas Partners — **Remote, USA**
Machine Learning Engineer — *May 2021 - November 2022*

- **Stock Movement Classification**: Implemented a K-Nearest Neighbours classifier to predict directional stock movements, using Pandas, NumPy, and scikit-learn for feature engineering and preprocessing.
- **Deep Learning for Time-Series**: Developed LSTM and AutoEncoder architectures to model financial time-series data, capturing non-linear market dependencies beyond traditional statistical approaches.
- **Paper-Trading Pipeline**: Built an automated paper-trading simulation framework using Backtrader

to validate ML strategy performance against historical market data.

Afiniti **Lahore, Pakistan**

Research Analyst / iOS Developer *August 2020 - May 2021*

- **Performance Optimisation**: Refactored the Confu conference audio synchronisation app, replacing nested loop operations with vectorised matrix operations via Apple's Accelerate framework, significantly reducing runtime and CPU load.
- **Numerical Computing**: Implemented matrix exponential computation via eigendecomposition using LAPACK, providing a custom numerical solution absent from Apple's Accelerate framework.
- **SwiftSpeechFeatures**: Ported the CSpeechFeatures library in full from C programming language to Swift, maintaining numerical precision throughout.
- **iOS Frontend**: Built the SwiftUI frontend for Confu, implementing Adobe XD designs pixel-perfectly and managing audio state flow across multi-device conference sessions.

OMNO AI **Lahore, Pakistan**

Computer Vision & Deep Learning Engineer *March 2019 - August 2020*

- **Multi-Task Inference Pipeline**: Developed a real-time system combining YOLO for person detection, Haar Cascade for face localisation within YOLO bounding boxes, and 3 independent neural networks for facial emotion recognition, age estimation, and gender classification on live IP camera footage.
- **Sports Analytics**: Designed a K-Means clustering pipeline to classify soccer players by jersey colour in broadcast footage, enabling automated team identification without labelled training data; also fine-tuned YOLO for real-time soccer ball tracking.
- **Edge & Infrastructure**: Configured and maintained on-premise Linux GPU servers as the primary deep learning training environment; also evaluated model inference on Raspberry Pi with Intel Neural Compute Stick for resource-constrained edge scenarios.

PERSONAL PROJECTS

- **LLM from Scratch**: Implementing transformer components from the ground up following Sebastian Raschka's *Build a Large Language Model*; running workloads on a custom PyTorch ROCm environment built around an AMD RX 9070 XT. Active.
- **Portfolio Website**: Rebuilt on the Astro framework, migrating from a vanilla HTML/CSS/JS codebase to a Markdown-driven content pipeline for future project write-ups; deployed via GitHub Pages at ibtihajt.github.io.
- **Diagnostic ECG Machine**: Acquired an AD8232 analogue front-end and Arduino Uno to capture and visualise live Lead I ECG signals; obtained the PTB-XL clinical ECG dataset as the training foundation for a planned AI-based cardiac diagnostic model. Hardware prototype functional; AI development stage pending.
- **CNN Filter Visualisation on MNIST**: Visualised the learned convolutional filter activations of CNNs trained on MNIST as a personal exploration of deep learning interpretability; public at github.com/IbtihajT/Visualization-on-the-MNIST.

EDUCATION

University of Chester **Chester, England**

MSc Advanced Computer Science *October 2022 - October 2023*

- **Dissertation (Distinction)**: Singing Emotion Prediction with Machine Learning — Trained 6 models (Logistic Regression, SVC, Nu-SVC, Random Forest, Extra Trees, DNN) for 6-class emotion classification from audio features (Spectral, Temporal, HNR, MFCC, Prosodic), achieving a median F1-Score of 0.85, with the best model reaching F1 = 0.90.
- **Robotics & BCI**: Developed a conceptual brain-controlled prosthetic hand interface using EEG signals and Python (Webots simulation).
- **Advanced Algorithms**: Implemented Memetic Algorithms and Hill Climbing heuristics to solve NP-hard combinatorial optimisation problems.

FAST-NUCES **Lahore, Pakistan**

BSc Computer Science (Hons) *January 2013 - January 2019*

- **Specialisation**: Machine Learning, Deep Learning & Computer Vision.
- **Final Year Project**: Malware Analysis via Bytecode Visualisation — Applied CNNs to extract spatial features of malicious code by converting binary files into image representations.
- **Coursework**: Bio Informatics — developed a CNN for human gene classification from DNA sequences (course project).